

Total No. of Questions : 8]

SEAT No. :

PE-2486

[Total No. of Pages : 2

[6583]-10

T.E. (Civil Engineering)

**HYDROLOGY AND WATER RESOURCES ENGINEERING
(2019 Pattern) (Semester - V) (301001)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.No.6, No.7 or Q.No.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) What are the various methods to estimate flood and explain any one method in detail. **[9]**

b) Explain briefly the terms: **[8]**

- i) Design flood.
- ii) Standard project flood
- iii) Probable maximum flood
- iv) Design storm

OR

Q2) a) What is flood routing, types of flood routing and explain Muskingum method of flood routing. **[10]**

b) What are the application of software in hydrology and what are the different hydrological software commonly used. Explain application Q-GIS software in watershed delineation. **[7]**

Q3) a) What are the different types of reservoir explain one reservoir in each type? **[8]**

b) List out the various zones of storage in a reservoir. Explain this with a neat sketch, the sedimentation zones in reservoir and methods to control sedimentation of reservoir. **[9]**

OR

P.T.O.

- Q4) a)** Explain yield and different types of yield from a reservoir. Explain 'mass curve analysis' for determining the capacity of a reservoir for a specified yield. [9]
- b)** Explain density currents, trap efficiency, useful life of reservoir, reservoir losses. [8]

- Q5) a)** What is water logging, state any four causes of water logging. What are the ill effects of water logging and how it is controlled. [8]
- b)** State Dupit's assumptions and derive the equation for discharge from unconfined aquifer under steady state condition. [10]

OR

- Q6) a)** Differentiate between aquifer, aquiclude, aquitard and aquifuge with suitable example. [8]
- b)** A well of 300 mm diameter is located in a confined aquifer of 40m thick. The aquifer has a hydraulic conductivity of 25 m/day and the radius of influence is 300m. determine the discharge in m^3/day from the well if the draw down is 3m. Also determine the percentage increase in discharge if the diameter is made to 450mm, with the other conditions remain same. [5]
- c)** How does the permeability of aquifer affect the groundwater movement? Explain the Darcy equation governing the flow of water through aquifer medium. What are the limitations of Darcy equation? [5]

- Q7) a)** What is irrigation? Why is it necessary? What are the benefits and ill effects of irrigation of irrigation projects? [8]
- b)** Explain the following terms [10]
- Crop period
 - Base period
 - Intensity of irrigation
 - Paleo irrigation
 - Consumptive use

OR

- Q8) a)** Explain the significance of Duty and delta in irrigation and how duty varies with the place of measurement. Develop the relation between delta, base period and duty for crop. [8]
- b)** State concept of PIM and explain role of water user's cooperative society in PIM. What are the merits and demerits of area or crop basis and volumetric assessment of water charges? [10]

